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Visualization of crowd contamination simulations using Immersive Virtual Reality

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***Abstract*—Smart cities can generate a lot of useful data for analyzing behavior and infrastructures, helping with urban planning. However, the applications found are often limited to desktop viewing only, which can hinder or restrict data analysis. This work proposes an immersive data visualization of the population of medical centers to assist in the analysis and development of new strategies to deal with overcrowding situations. The prototype allows the visualization of crowd data located in places of interest on top of a 3D map. The user can view the number of people present at the location at the current stage of the simulation or over time. To test the prototype, data from an infectious disease simulator was used, called LODUS. The results showed that the application has potential for visualizing this type of data.**

***Keywords*—smart cities, simulation, visualization, virtual reality**

I. Introduction

Smart cities are urban centers capable of collecting data on various aspects of the population and generating improvements in the functioning of the city. For example, this data can help create strategies to avoid overcrowding in medical centers during an epidemic, such as Covid-19 [1].

Studies conclude that the use of smart cities and simulation data can improve urban planning and, consequently, people's quality of life [2, 3, 4]. In particular, simulations that allow the integration of real data to generate the behavior of groups of people, like LODUS [5], can significantly impact this type of analysis.

However, most analysis tools or simulators use dashboards to visualize this data, which sometimes may not be enough to analyze so much data or generate insights to solve the proposed problem [5, 6, 7]. Added to this is the fact that many simulators do not allow the parameters of the simulator to be edited at runtime, a mechanism that could be useful in the process of analyzing and developing strategies to solve the proposed problem.

Virtual Reality (VR) is a technology that can facilitate data visualization and analysis [8], as it allows 3D elements to be visualized and manipulated in an immersive environment more easily and intuitively, such as using controls to pick up and drop objects. Added to this is the engagement generated by VR applications [9], which can help gain new insights into the problem and assist in the development of the solution.

Based on this scenario, this work developed an application for immersive data visualization on the population of medical centers to help in the analysis and development of new strategies. We utilize the LODUS framework [5] to run an infectious disease simulation in a virtual environment and obtain data of the population.

The application allows you to visualize data from crowd simulations, monitoring the number of people concentrated in specific locations and informing you when a location has reached its maximum capacity. The application also allows the visualization of the number of people in the places at the current moment or over time, making it possible to analyze the data generated and possibly generate new insights into the problem.

II. Related Work

Most applications for visualizing simulation data only use 2D dashboards to display this data.

Franke et al. [10] developed a platform for crowd management in smart cities based on mobile sensors. Crowd data is displayed using a heatmap, showing the most populated areas. The platform also allows you to visualize the state of the crowds over time, using a slider to manage the timeline.

Following this, Atta et al. [11] created a framework for Big Data visualization, in which data is filtered and summarized after acquisition and displayed to the user in dashboard format. The framework provides a variety of visual resources and tools for summarizing crowds, such as the visualization of cluster markers, density, and heatmap, all in 2D.

Another work that uses crowd visualization is by Lopez-Novoa et al. [12]. In it, they created an overcrowding detection algorithm based on Wi-Fi requests and visualized this data on a platform that provides dashboards with summary data. The visualizations used are also crowd markers, informing the number of people detected in the area and the average density calculated in small sectors of the location.

Finally, the work by Silva et al. [5] develops a framework for a crowd simulator capable of generating an approximation of the dynamics of groups of people based on real data. This simulator generates discrete stages with the population variations in each region of interest defined. This data is also visualized using markers positioned on top of the map of the selected regions. This simulator has a 3D visualization option, but the interactions remain in 2D.

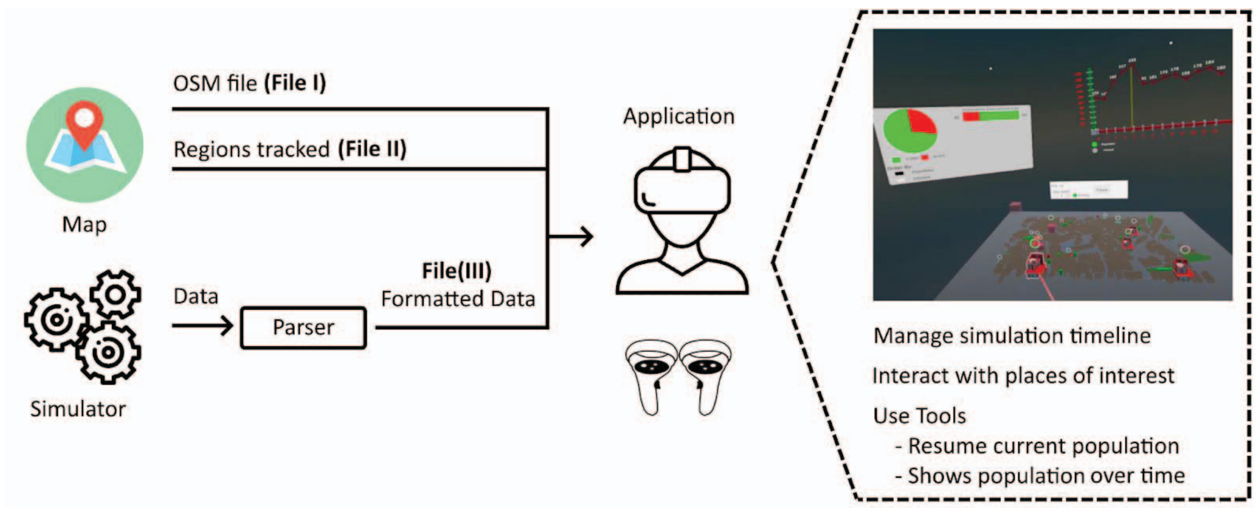

Fig. 1: Application overview

None of the available tools offer an immersive 3D visualization option, which could generate results that are easier to understand and interact with. For this reason, this work aims to implement the visualization and interaction of crowd data using an immersive virtual environment.

III. Application

The application developed in this work makes it possible to visualize and interact with simulation data using an immersive virtual environment. The system was implemented in Unity3D for the Meta Quest 2 platform.

Fig. 1 shows an overview of the application. The application receives three input files, generates the virtual environment with the 3D map, and displays the interfaces available for the user to manage the visualization of the simulation. The following sections describe each process in detail.

The application needs three files to load the environment. They are:

I. A file with the map region in which the data was generated.
II. A file with the regions to be analyzed.
III. A file with the simulation data formatted for the application.

File (I) is of the OpenStreetMap (OSM) [13] type, which maps all the locations, establishments, streets, and roads that will be used to visualize the simulation. This file can be generated using OSM's own API and contains the geographical locations (Latitude and Longitude) and geometry of the buildings in the region.

File (II) contains the list of nodes that will take part in the visualization. The file format is shown in Fig. 2. Each line of the file is structured as follows:

- **ID:** Identifier of the node.
- **Latitude:** The latitude of the node.
- **Longitude:** The longitude of the node.
- **Description:** Name of the building.
- **Type:** Type of building (Hospital, Normal commercial establishment, Church, Hotel, Petrol station).
- **Capacity:** Maximum number of people supported by the building.

```
Frame,Hour,Day,Node Origin,Node Destiny,Total,Infected
0,0,0,node1,node1,30,0
0,0,0,node2,node2,203,14
0,0,0,node3,node3,20,0
0,0,0,node4,node4,200,60
0,0,0,node5,node5,200,120
0,0,0,node6,node6,200,75
0,0,0,node7,node7,15,0
0,0,0,node8,node8,30,0
0,0,0,node9,node9,23,0
0,0,0,node10,node10,12,0
0,0,0,node11,node11,7,0
0,0,0,node12,node12,3,0
0,0,0,node13,node13,70,0
1,1,0,node1,node2,7,2
2,2,0,node2,node3,6,1
3,3,0,node3,node2,10,7
4,4,0,node4,node2,5,0
5,5,0,node5,node6,9,4
6,6,0,node6,node1,10,1
7,7,0,node1,node5,7,2
8,8,0,node1,node3,8,1
9,9,0,node2,node5,3,7
10,10,0,node2,node6,5,0
11,11,0,node3,node1,9,4
12,12,0,node4,node1,4,1
```

Fig. 2: Application node file format.

First, confirm that you have the correct template for your paper size. This template has been tailored for output on the US-letter paper size. If you are using A4-sized paper, please close this file and download the file “MSW_A4_format”.

Finally, File (III) File III contains the data generated from LODUS. The data has been adapted to allow visualization of the path of each crowd displacement.. The file format is shown in Fig. 3. The columns represent:

- **Frame:** A time unit.
- **Hour/Day:** Time identifier (optional).

- **Node Origin:** The source region in which an event took place.
- **Node Destiny:** The destination region in which an event took place.
- **Total:** Total number of people present in the region at the end of the step.
- **Infected:** Total number of infected people in the region at the end of the step.

```
node1,-51.215430,-30.025619,Posto Ipiranga,-1,100
node2,-51.214619,-30.029496,Hospital Materno Infantil Presidente Vargas,0,250
node3,-51.212920,-30.027816,Poza Café,-1,100
node4,-51.206690,-30.029217,Hospital Fêmina,0,250
node5,-51.209060,-30.026692,Emergência do Hospital Moinhos de Vento,0,250
node6,-51.208311,-30.024851,Hospital Moinhos de Vento,0,250
node7,-51.217125,-30.028762,Restaurante Barros Cassal,-1,100
node8,-51.214839,-30.027286,Pousada & Hostel Bahia,-1,100
node9,-51.215933,-30.025435,Lagoinha Porto Alegre,-1,100
node10,-51.214479,-30.024731,Lótus Hotel e Hostel,-1,100
node11,-51.213804,-30.023415,SuperAuto Ford,-1,100
node12,-51.210119,-30.022045,Molduras Santos,-1,100
node13,-51.209191,-30.022998,Igreja Adventista do Sétimo Dia - Moinhos,-1,100
```

Fig. 3: Format of the application's event file.

A node represents a building that will be monitored during data visualization. The nodes are represented as more elaborate 3D models than the other buildings on the 3D map. The first frame of the file is used to start the population in the participating nodes.

This separation of files (participating nodes and event data) allows the user to select only the regions of interest to view without generating a simulator file for each group of regions. The capacity of the building is used so that when the venue has more people than it can handle, a signal appears that the venue is at maximum capacity, highlighting the regions with overcrowding problems and potentially generating insight when analyzing the data.

A. 3D Map Generation

The 3D map is created using an open source project from a Github profile called SnakeSneakS [14]. Here, the OSM file is loaded and the information from the nodes is used to construct the shape of the roads and buildings. Fig. 4 shows an example of a map generated by the project.

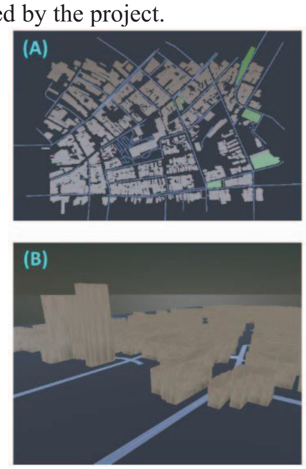

Fig. 4: Example of a 3D map created in the application. Part (A) shows a top view of the generated map and (B) shows the buildings generated in 3D.

The generation of buildings uses the site' geometry to create a mesh with 3D points from the virtual environment, but with the same value on the y-axis, resulting in a 2D geometry figure. Once the mesh is complete, this figure is projected to a predefined height and the faces are calculated, generating a 3D model of the building.

B. Simulation Steps

In LODUS, each simulation step contains a list of the regions used and the number of healthy and infected people in the current step. To display the flow of people in and out of the locations represented by this data, the application adds a random origin or destination region just to demonstrate the flow of people.

To create this animation, a route between the two regions must be calculated. This calculation is done using a route calculation API in an OSM file, called OSRM [15]. For each region in the simulation step, the route between the region and another random region is calculated and stored in an NxN matrix, where N is the total number of regions in the simulation. This matrix is used to avoid the unnecessary network and processing overhead of recalculating routes that have already been requested.

After all routes are calculated, the application creates the 3D map and adds the models for the regions of interest. Finally, after all the virtual scenarios are created, the simulation interface is displayed to the user, who can start visualizing the data.

C. Simulation Control

The simulation is controlled via joysticks interacting with a 2D interface and the 3D elements added to the scene. Fig. 5 shows the elements present in the environment for interaction.

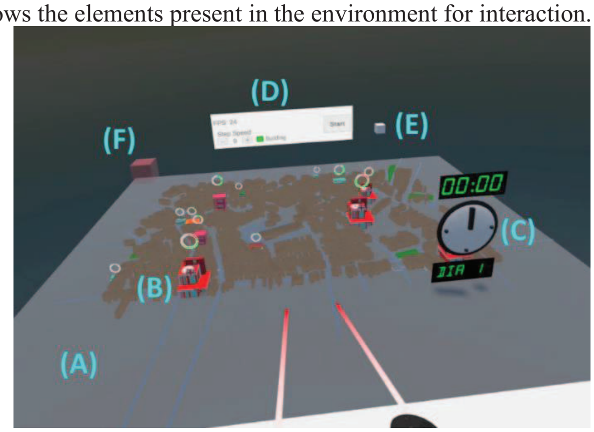

Fig. 5: Scene elements during visualization of simulation steps: (A) 3D map created from file; (B) 3D model used to differentiate types of establishments; (C) simulation timeline controller; (D) user interface to start simulation; (E) line graph; and (F) the selection tool.

The 3D map (Fig. 5A) is displayed on top of a plane that can be moved using the controls. The map created displays the streets, roads, buildings, and green areas of the region selected in the configuration file. If the name of a building is present in the OSM file, the user can check it by pointing the control beam at it

For the buildings that will take part in the simulation (Fig. 5B), a 3D model is instantiated based on the type of establishment selected. All the models used for this purpose were taken from a free Unity Asset [16]. To display more information in a non-textual way, the models also have a circle that informs you of the building's current capacity, showing a warning sign when it reaches its maximum capacity. Fig. 6 shows an example of this indicator. As the building gets fuller, the size of the 3D model also increases.

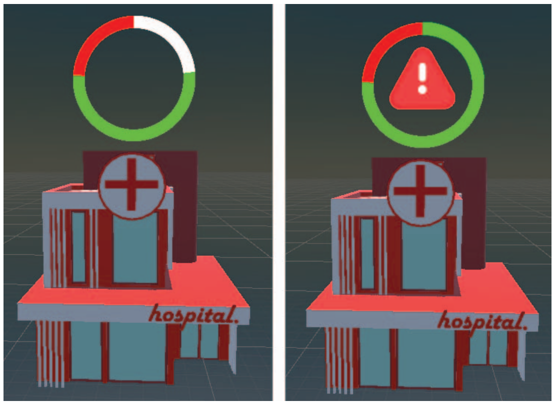

Fig. 6: Example of a building with a capacity indicator.

The user can navigate through the stages of the simulation by turning the clock hand (Fig. 5C) clockwise to move forward through the stages, or counterclockwise to move backwards through the stages. The monitor above the clock displays the animation time between steps. By accessing the 2D interface (Fig. 5D), users can start the visualization, change the speed of the steps, or turn off the visualization of unused buildings.

The application also has two tools to help visualize the data: the line graph (Fig. 5E) and the area selection (Fig. 5F). The line graph (Fig. 5E) displays the number of people present in the buildings selected with the control. For each building selected, a line is added to the graph, which is highlighted if the control beam is pointed at the target building. Fig. 7 shows an example of the tool. The current population data for each building can also be viewed by a 2D monitor inserted into the scene in which at least one building is selected (Fig. 8).

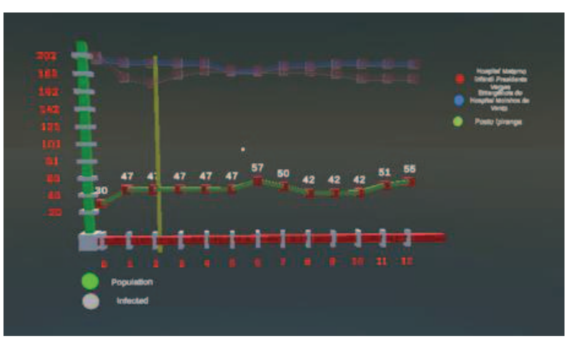

Fig. 7: Line graph showing the capacity of the monitored buildings.

Fig. 8: Population monitor during the simulation step.

Finally, the group selection tool (Fig. 5F) allows you to select all the regions that collide with the semi-transparent parallelepiped shown in Fig. 9. The selected buildings are added to the 2D monitor and the line graph (Fig. 5E). To remove a building from the selection, the user needs to move the tool so that it does not collide with the building or just select it again with the control.

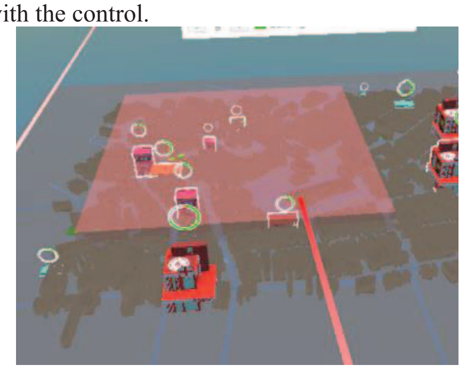

Fig. 9: Example of using the group selection tool.

D. Data Visualization

The simulation steps are displayed when the user selects the Start button. The first step starts at this point, and the displacement events are instantiated on the map. Each displacement is represented by groups of "people" (cylinders) that move from the source node to the destination node at random speed. Green cylinders represent healthy people and red cylinders represent infected people, as shown in Fig. 10.

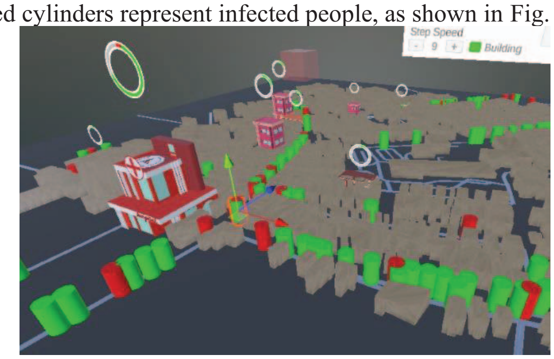

Fig. 10: Example of the crowd representation used in the Application.

This display is merely illustrative, as the aim is to show how the number of people in each building varies over time. As the simulation data progresses, the clock hand is rotated clockwise, simulating the passage of time.

E. Application Test

To test the prototype developed, we used a configuration file with 13 regions of interest, located in the Independência neighborhood of Porto Alegre, Brazil, and seven simulation steps with randomly generated events.

Fig. 11 shows a frame from the demonstration. It is possible to see the displacement events represented by groups of cylinders departing from a source node to a destination. The green cylinders represent healthy people, while the red cylinders represent infected people. The Clock is updated every frame, in which the minutes represent the animation time of the step and the hours represent the current step. A full demonstration is available on Youtube http://tiny.cc/DemoVideo_CCVR [17].

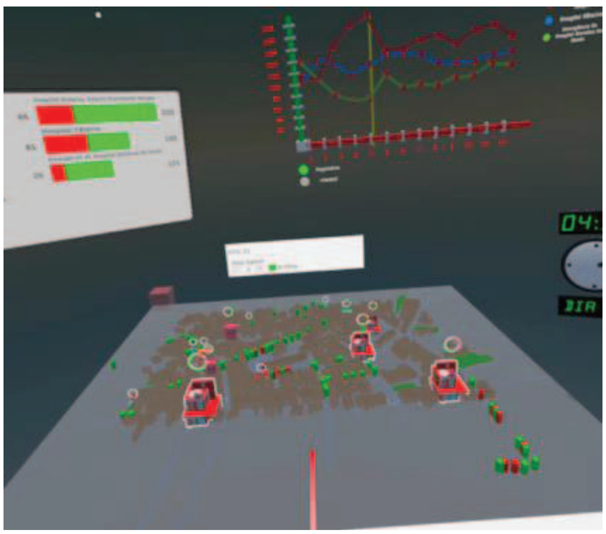

Fig. 11: Example of data visualization in the application.

IV. CONCLUSION

In this work, we have implemented a prototype for visualizing data from crowd contamination simulations in an immersive virtual reality environment. The prototype allows a 3D map region to be loaded and regions of interest to be instantiated to monitor people's entry and exit. A region grouping tool was created to allow the user to visualize the data of a specific region of the map, enabling a better understanding of how buildings deal with overcrowding and allowing insights into strategies to improve this process.

Overall, the prototype met the necessary requirements to allow the visualization of data in an immersive way. It lets the user select the regions of interest and create a summary of the information contained in the selection. User studies should be carried out to validate the usability and performance of the application for analyzing data and planning new approaches.

We consider our main contribution to be the way we visualize and interact with simulator data in an immersive virtual reality environment. We believe that the visualization of the 3D map with the interaction using the controls and data summary tools can help in the analysis of the data and generate insights to solve the proposed problem.

V. ACKNOWLEDGEMENT

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